Department of Electrical and Computer Engineering
Summer Semester, 2025/2026
Digital Integrated Circuits - ENCS3330
Project, August 21, 2026
Due Date: September 10, 2026

Low-Power CMOS D Flip-Flop and 4-bit Register/Counter

Sequential-cell design, timing characterization, layout, and low-power clocked-system evaluation

Course	Digital Integrated Circuits
Main tools	Electric VLSI for schematic/layout; LTspice for transistor-level simulation
Group size	3 students per group
Core methodology	Baseline design -> optimized design -> schematic simulation -> layout -> quantitative comparison
Final deliverables	Electric files, LTspice package, technical report, comparison tables, and presentation

Project philosophy: Master the complete custom digital-IC flow using small reusable CMOS blocks, then integrate them into a larger circuit and defend design choices using measured area, delay, power, and robustness. **Important:** The project is an educational design exercise. Results should be identified as schematic simulation, layout-based estimates, or extracted simulation; do not describe them as fabricated-silicon measurements.

1. Project Overview

Design a conventional CMOS D flip-flop (DFF), develop or select an instructor-approved lower-power alternative, and integrate the selected cell into a 4-bit register and a 4-bit counter. The project emphasizes the clocked storage element as a real IC cell: correct functionality, setup/hold behavior, clock-to-Q delay, power, transistor count, layout area, and system-level consequences.

2. Learning Objectives

- Design CMOS latches and edge-triggered D flip-flops at transistor level.
- Understand transmission gates, clock inversion, master-slave operation, and internal clock loading.
- Measure clock-to-Q delay, setup time, hold time, rise/fall time, average power, and PDP.
- Compare a conventional DFF against a lower-power or lower-transistor-count alternative.
- Create reusable Electric layouts and integrate repeated DFF cells into a 4-bit sequential block.
- Evaluate how clock frequency, data activity, VDD, and load affect energy consumption.

3. Required Circuit Blocks

1. CMOS inverter and transmission gate.
2. Level-sensitive D latch.
3. Baseline positive-edge-triggered DFF, preferably transmission-gate master-slave.

4. Optimized DFF: instructor-approved lower-power alternative (for example clock-gated, reduced internal switching, or another static/semi-static topology appropriate to the course).
5. 4-bit parallel register using the selected DFF.
6. 4-bit binary counter. A synchronous counter is preferred for excellent work; a ripple counter may be used as a baseline/comparison.

4. Design Challenge

Baseline: conventional transmission-gate master-slave DFF. Proposed design: reduce energy or PDP while preserving correct state storage and acceptable timing. The group must explain where switching activity or clock loading is reduced and verify the claim with LTspice measurements.

- Do not optimize only transistor count. Verify full logic levels, correct edge triggering, timing margins, and load drive.
- If clock gating is used, it must be implemented and tested carefully so that the group can explain its effect on switching activity.
- All comparisons must use identical device models, VDD, clock frequency, load, and data patterns.

5. Required Timing and Power Characterization

Metric	**Required method / evidence**
Functionality	Show Q sampling D on the specified clock edge and retaining state between edges.
Clock-to-Q, tCQ	Measure from the active clock 50% crossing to Q 50% crossing.
Setup time	Sweep the D transition toward the active clock edge and identify the boundary used by the group.
Hold time	Sweep D after the active edge and identify the boundary used by the group.
Power	Average supply power over a stated clock/data sequence; include clock activity.
PDP / energy	Report a consistent power-delay or energy-per-cycle metric.
Robustness	Repeat selected tests versus VDD, clock frequency, and output load.

6. Register and Counter Integration

Use four instances of the selected DFF to create a 4-bit parallel register. Then implement a 4-bit binary counter and verify the full counting sequence.

1. Register: demonstrate load/capture and stored outputs Q3:Q0.
2. Counter: demonstrate a complete count cycle and reset behavior if implemented.
3. Measure maximum observed propagation delay from the active clock edge to the final output transition.
4. Compare power for at least two data-activity conditions (for example low activity versus frequent toggling).
5. For a ripple-counter baseline, observe accumulated stage delay. For a synchronous design, discuss the added combinational logic versus timing benefit.

7. Electric VLSI Requirements

- Create hierarchical schematic/symbol/layout cells for inverter, transmission gate, latch, baseline DFF, optimized DFF, and repeated sequential blocks.
- Use consistent VDD/GND rails and a layout strategy that allows DFF cells to be tiled into the 4-bit register/counter.
- Pay special attention to clock routing, clock inverter placement, and repeated-cell organization.
- Run DRC where available; document unresolved issues.
- Create extracted/post-layout simulation for at least the DFF if supported by the available flow.

8. LTspice Simulation Plan

1. Verify inverter and transmission-gate behavior.
2. Verify the latch in transparent and hold modes.
3. Verify the baseline DFF for all relevant D/CLK transitions.
4. Measure t_{CQ} and estimate setup/hold using a documented sweep procedure.
5. Measure average power for a fixed clock/data sequence.
6. Repeat for the optimized DFF under identical conditions.
7. Integrate four DFFs into the register and counter and verify sequential operation.
8. Run selected VDD, load, and clock-frequency sweeps and summarize trends.

9. Minimum, Excellent, and Bonus Expectations

Level	**Expected evidence**
Minimum passing	Working latch and DFF, functional LTspice waveforms, one DFF layout, and working 4-bit register or counter.
Excellent	Baseline + optimized DFF comparison, setup/hold/t_{CQ} characterization, register and counter, DRC-aware repeated-cell layout, VDD/frequency/load sweeps.
Bonus	Post-layout/extracted timing, synchronous-versus-ripple counter comparison, clock-gating analysis, energy per useful operation, or Monte Carlo/process variation if models permit.

10. Deliverables

Deliverable	**Required content**
Electric package	Hierarchical schematics/symbols/layouts, library file, DRC notes, extracted netlist if available.
LTspice package	Testbenches, clock/data patterns, timing sweeps, power measurements, VDD/frequency/load sweeps.
Technical report	Abstract, related work, baseline/proposed DFF, timing methodology, layout, register/counter, results, limitations, conclusion.
Comparison table	Transistors, area, tCQ, setup, hold, average power, PDP/energy, operating conditions.

Deliverable	Required content
Presentation	8-12 slides with architecture, transistor schematics, layouts, timing diagrams, comparison table, contribution and limitations.

11. Evaluation Rubric (100 points)

Category	Points	Excellent performance
Technical understanding	10	Explains latch/DFF operation, clocking, timing constraints and power mechanisms.
Baseline DFF	10	Correct transistor-level design and clean functional behavior.
Optimized DFF	10	Technically justified low-power improvement with correct operation.
Timing characterization	10	Clear and consistent tCQ/setup/hold methodology and results.
Power/PDP analysis	10	Fair power comparison with stated clock/data activity.
Electric layout	10	Compact, reusable, DRC-aware DFF and sequential-block layout.
4-bit integration	10	Correct register and counter operation with timing discussion.
Results/report	10	Professional evidence-based comparison, figures, tables, citations, limitations.
Presentation/defense	20	All members can defend the complete project.

AI tools may assist explanation or debugging only; students remain responsible for every design and results.